

SREDNY VASYUGAN, Russian Federation

WMO: 291110

Lat: 59.217N

Lon: 78.233E

Elev: 69

StdP: 100.50

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.4	(c) -36.0	(d) -41.9	(e) 0.1	(f) -39.0	(g) -38.5	(h) 0.1	(i) -35.8	(j) 6.8	(k) -9.3	(l) 6.2	(m) -11.5	(n) 0.7	(o) 50	(p) 0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	28.7	18.9	27.1	18.5	25.3	17.8	20.6	26.1	19.6	24.9	18.6	23.7	2.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.6	23.4	17.6	12.7	22.2	16.6	11.9	21.2	59.8	26.1	56.3	24.9	53.2	23.7	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.7	5.9	5.3	-41.0	31.5	4.3	1.5	-44.1	32.6	-46.6	33.4	-49.1	34.3	-52.2	35.3
			-41.3	22.3	3.9	1.4	-44.1	23.3	-46.4	24.2	-48.7	24.9	-51.5	26.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-20.0	-16.4	-6.6	0.8	8.5	16.2	18.3	14.8	8.5	0.9	-10.5	-16.9
	DBStd	14.73	9.06	8.21	6.64	6.17	5.83	5.15	3.71	3.84	4.58	5.29	8.87	9.23
	HDD10.0	4391	929	740	515	278	102	11	0	4	81	283	615	834
	HDD18.3	6826	1187	973	773	525	309	99	47	121	295	539	865	1092
	CDD10.0	707	0	0	0	3	54	197	259	155	37	2	0	0
	CDD18.3	101	0	0	0	0	4	36	48	13	1	0	0	0
	CDH23.3	843	0	0	0	0	45	304	382	107	6	0	0	0
	CDH26.7	190	0	0	0	0	11	78	85	17	0	0	0	0
	WSAvg	2.4	2.2	2.4	2.8	2.8	2.8	2.2	1.9	2.0	2.3	2.6	2.6	2.5
	PrecAvg	550	29	20	20	32	53	74	72	82	51	48	41	32
(15) Precipitation	PrecMax	746	48	69	50	80	129	181	185	155	99	109	78	61
	PrecMin	385	8	5	2	8	16	5	8	0	13	18	14	4
	PrecStd	97	10	12	11	18	26	35	37	41	22	21	16	13
	PrecStd	97	10	12	11	18	26	35	37	41	22	21	16	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.1	0.7	8.3	19.1	28.3	31.1	31.0	28.6	23.6	15.4	4.7	0.5
		MCWB	-0.6	-0.6	3.3	10.5	16.5	18.6	21.0	19.5	14.9	10.1	3.4	-0.1
	2%	DB	-2.7	-1.1	5.3	14.8	24.1	28.9	28.7	26.2	20.7	12.4	2.6	-1.1
		MCWB	-3.4	-2.1	1.2	8.2	14.1	18.4	19.4	18.5	14.5	8.2	1.7	-1.7
	5%	DB	-5.3	-3.1	3.2	12.1	21.2	26.8	27.3	24.0	18.2	10.4	1.3	-2.6
		MCWB	-6.0	-4.0	0.3	6.6	12.9	17.9	19.2	17.6	13.2	7.4	0.5	-3.2
	10%	DB	-7.6	-5.4	1.7	9.7	18.1	24.8	25.5	21.7	15.8	8.2	0.2	-4.6
		MCWB	-8.2	-6.4	-0.4	5.1	11.5	17.3	18.4	16.7	12.1	6.1	-0.4	-5.2
	0.4%	WB	-0.5	-0.1	3.5	11.2	17.2	21.5	22.8	20.9	16.8	10.7	3.5	0.0
		MCDB	0.1	0.5	7.6	18.3	26.7	27.8	28.9	26.5	21.1	13.5	4.3	0.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-3.4	-2.0	1.8	8.9	15.1	19.9	21.2	19.4	15.1	9.1	1.8	-1.7
		MCDB	-2.8	-1.2	4.1	14.3	22.1	26.0	26.7	24.4	19.4	11.7	2.4	-1.2
	5%	WB	-5.9	-4.0	0.7	7.2	13.6	18.9	20.1	18.3	13.7	7.7	0.7	-3.2
		MCDB	-5.4	-3.2	2.8	11.1	19.7	24.9	25.1	22.6	17.6	10.0	1.2	-2.6
	10%	WB	-8.2	-6.4	-0.3	5.4	11.9	17.8	19.1	17.2	12.3	6.1	-0.4	-5.2
		MCDB	-7.7	-5.4	1.8	9.3	17.6	23.2	24.1	21.2	15.4	8.1	0.2	-4.6
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	8.2	9.2	9.6	9.5	9.3	9.6	8.9	7.6	5.4	6.0	6.8
		MCDBR	6.6	6.3	9.4	12.2	14.0	12.5	12.5	12.8	11.7	8.1	5.2	7.1
		MCWBR	6.5	5.8	6.6	7.3	6.6	4.9	5.3	6.1	6.3	5.2	5.0	6.9
		MCDBR	6.6	6.1	7.8	11.3	12.4	11.0	10.7	11.0	10.3	7.6	4.8	6.9
	5% WB	MCWBR	6.5	5.7	5.9	7.3	6.7	5.0	5.3	5.9	6.0	5.2	4.8	6.8
		MCWBR	6.5	5.7	5.9	7.3	6.7	5.0	5.3	5.9	6.0	5.2	4.8	6.8
(40) Clear-Sky Solar Irradiance	taub		0.269	0.307	0.359	0.385	0.390	0.407	0.446	0.413	0.365	0.335	0.287	0.267
	taud		1.996	2.015	1.951	2.070	2.289	2.319	2.186	2.264	2.357	2.269	2.056	1.938
	Ebn at Noon		491	670	743	806	838	827	778	776	759	651	463	339
	Edh at Noon		53	88	128	136	118	116	130	112	87	69	49	39
(44) All-Sky Solar Radiation	RadAvg		0.54	1.49	2.97	4.36	4.87	5.55	5.07	3.74	2.38	1.13	0.49	0.30
	RadStd		0.06	0.12	0.21	0.42	0.52	0.51	0.52	0.41	0.31	0.10	0.11	0.03

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page